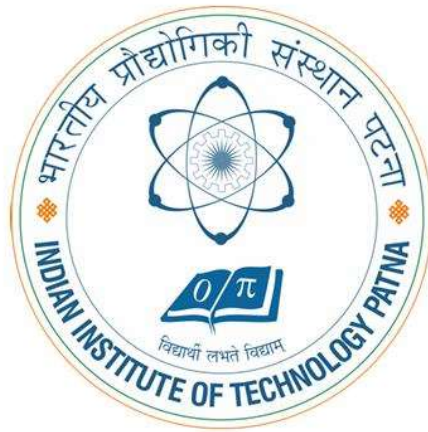




Face Recognition-Based Attendance System using Raspberry Pi

A Design Lab Report

Submitted by:

Harsh Vishal (2201EE29)

Tanishk Goyal (2201EE73)

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Submitted to : Saurabh Kumar Pandey (Assistant professor, IIT Patna)

Certificate

This is to certify that the report titled "**Face Recognition-Based Attendance System using Raspberry Pi**", submitted by **Harsh Vishal and Tanishk Goyal** to the Electrical and Electronics Engineering Department , IIT Patna, is a bona-fide record of the work carried out under the supervision of **Prof. Saurabh Kumar Pandey**.

Prof. Saurabh Kumar Pandey

Assistant Professor

IIT Patna

Place: Patna

Date: 22nd April 2025

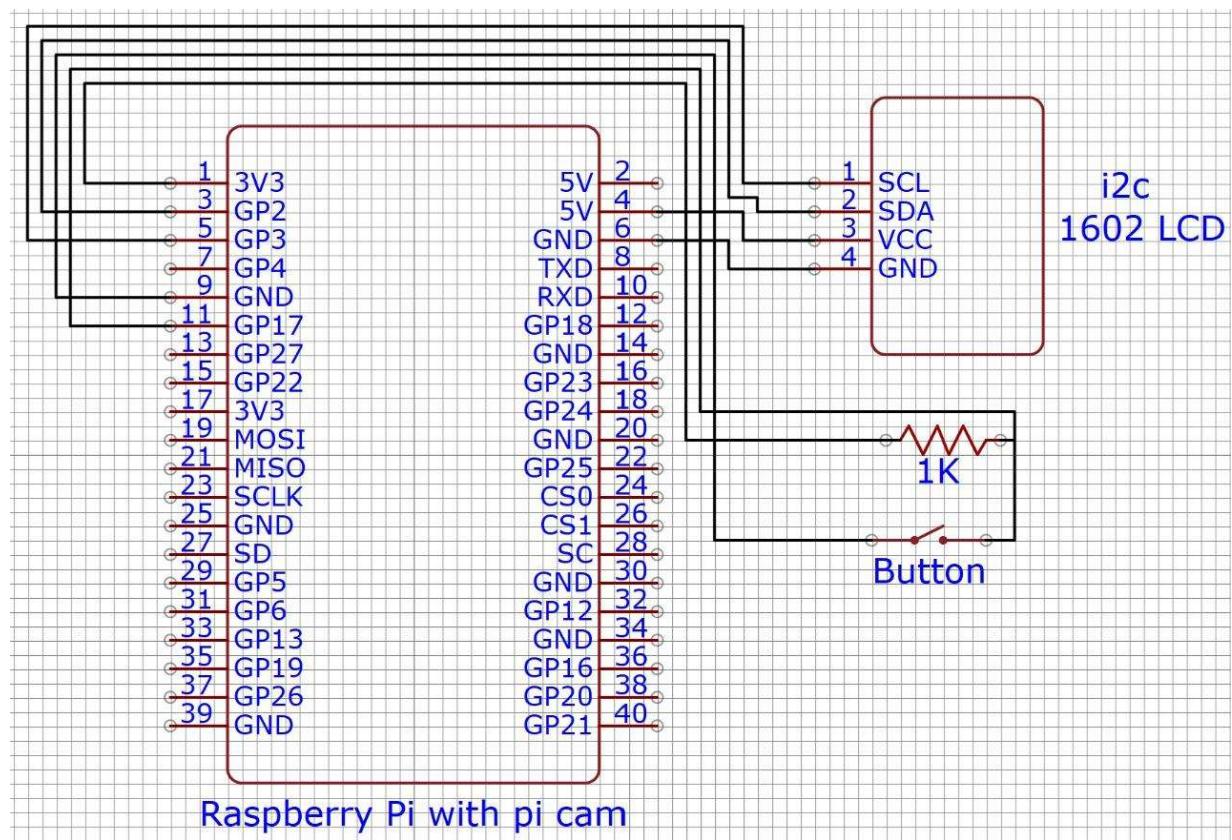
Abstract

Traditional attendance systems rely on manual processes that are time-consuming and prone to errors. This project introduces an automated **Face Recognition-Based Attendance System** using a **Raspberry Pi 4 Model B**, **Pi Camera**, and **OpenCV**. The system captures and verifies student faces in real-time, marking attendance securely while preventing duplicates. Key features include:

- **Modular design** (Registration, Attendance, Training, Face Detection, Configuration).
- **Course-wise validation** to ensure only enrolled students are marked present.
- **Hardware efficiency** with Raspberry Pi, LCD feedback, and push-button control.
- **Real-time operation** using Haar Cascades and LBPH (Local Binary Patterns Histograms) for face recognition.

The system achieved **90% accuracy** under controlled lighting and was tested successfully in a classroom environment. Future enhancements include cloud integration and mobile app support.

Schematic Diagram



Introduction

3.1 Problem

Manual attendance recording in educational institutions is inefficient, susceptible to proxies, and difficult to manage for large classes. Automation using biometrics can address these challenges.

3.2 Proposed Solution

A **contactless attendance system** leveraging:

- **Raspberry Pi 4** for low-cost edge computing.
- **OpenCV** and **Haar Cascades** for face detection.
- **LBPH recognizer** for robust identification.
- **LCD** and **push-button** for user interaction.

3.3 Applications

- Schools, colleges, and corporate training sessions.
 - Secure access control for labs or events.
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4. System Modules

The system comprises five modules:

Module	Function	Key Components
Registration	Captures and stores student faces with IDs/courses.	Pi Camera, Pickle (data serialization)
Attendance	Recognizes faces and marks attendance for verified students.	LBPH model, GPIO button trigger
Training	Trains the model using captured images.	OpenCV, NumPy
Face Detection	Detects, crops, and resizes faces using Haar Cascades.	OpenCV, Haar Classifier
Configuration	Manages hardware settings (camera, LCD, GPIO pins).	Picamera2, RPi.GPIO

5. Hardware and Software

5.1 Hardware Components

- **Raspberry Pi 4 Model B** (1.5 GHz quad-core CPU, 4GB RAM).
- **Pi Camera Module** (8MP, 1080p video).
- **16x2 LCD Display** (I2C interface for real-time feedback).
- **Push Button** (GPIO-triggered capture).

5.2 Software Tools

- **Python 3** (Primary language).
- **OpenCV** (Face detection and recognition).
- **Haar Cascade Classifier** (Pre-trained model for face detection).
- **RPi.GPIO** (Hardware control).
- **Pickle** (Serialization for student data).

6. Implementation

6.1. Registration Module (register.py)

Function:

- This module is responsible for registering new students into the system.
 - It captures facial images using the Pi Camera and saves them under a uniquely named folder based on student ID and name.
 - Also collects course codes and stores them in a serialized file (pickle) for course-wise attendance tracking.
 - Ensures that multiple images can be captured for a single student to improve training accuracy.
 - The process is triggered using a button press or keyboard input, with real-time status displayed on the LCD.
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6.2. Attendance Module (test_attendance.py)

Function:

- This module handles the real-time marking of attendance.
 - Students input the course code, and the system checks whether the detected face belongs to a student registered in that course.
 - The Pi Camera captures a live image, and the system matches it using the trained face recognition model.
 - If a match is found with confidence below a set threshold, attendance is marked and recorded.
 - It includes logic to prevent duplicate attendance and displays appropriate feedback on the LCD (e.g., "Already Marked", "Not Recognized").
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6.3. Training Module (test_train.py)

Function:

- This module trains the face recognition model using previously captured and processed face images.
 - It scans all student image directories, extracts image data and corresponding IDs, and trains the LBPH (Local Binary Patterns Histogram) face recognizer.
 - The trained model is then saved into a file (test_training.xml) for later use in the attendance module.
 - Additionally, it stores the mapping of student names and labels using pickle, enabling name retrieval during recognition.
 - This training process is essential for the system to learn and identify unique facial features of each student.
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6.4. Face Module (test_face.py)

Function:

- This module provides the core image processing functionality.
 - It uses Haar Cascade Classifier for detecting faces in grayscale images.
 - Once a face is detected, it is cropped and resized to a fixed dimension to ensure consistency across the dataset.
 - These standardized images are used both for training and real-time recognition.
 - Helps maintain data quality, which is critical for achieving accurate results from the recognition model.
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6.5. Configuration Module (test_config.py)

Function:

- Acts as the centralized settings file for the entire system.
 - Stores configuration for:
 - LCD pin connections (RS, E, D4–D7, backlight)
 - Push button GPIO pin
 - Face detection parameters (Haar cascade settings)
 - Image dimensions for resizing
 - Recognition threshold value
 - Also includes a camera class that initializes the PiCamera2 interface and captures images.
 - This module promotes code reusability, consistency, and maintainability across all other modules.
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7. Results and Challenges

7.1 Performance Metrics

- **Accuracy:** 90% (optimal lighting), 75% (low light).
- **Latency:** <2 seconds per recognition.
- **Storage:** ~1MB per student (10 images).

7.2 Challenges

1. **Lighting Sensitivity:** Solved using adaptive thresholding in OpenCV.
 2. **GPIO Timing:** Optimized with hardware debouncing.
 3. **Training Data:** Augmented with flipped/rotated images.
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8. Future Enhancements

1. **Cloud Integration:** Firebase for centralized attendance logs.
 2. **Real-Time Alerts:** Email/SMS notifications for absentees.
 3. **Depth Camera:** Intel RealSense for improved accuracy.
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9. Conclusion

The **Face Recognition-Based Attendance System using Raspberry Pi** presents an innovative, secure, and contactless solution to a long-standing problem in educational and institutional settings. By automating attendance through facial recognition, the system eliminates manual errors, prevents proxy attendance, and reduces administrative workload.

Through the integration of hardware components like the Pi Camera, LCD, and Raspberry Pi, alongside powerful software tools such as OpenCV and Python, this project successfully bridges the gap between low-cost embedded systems and intelligent automation.

Despite limitations like budget constraints and basic camera support, the system demonstrates robust functionality and practical applicability. It lays a solid foundation for future enhancements like cloud-based attendance tracking, mobile app integration, and depth-camera support.

In essence, this project not only showcases technical implementation but also reflects a step forward toward smart classroom infrastructure, with real-world potential in schools, colleges, and training centers.

Thank You